The Finnish paper reported a study of 24dB/octave crossovers at 100Hz and at 200Hz, with an adjustable delay unit switchable between the satellites and the subwoofers. It seems that delay is more detectable on speech than music, for starters. Also, it was found that the amount of tolerable delay was roughly inversely proportional to the crossover frequency. With a 100Hz crossover, there was an audible change in the sound by adding a 15ms delay to the satellites (placing the subwoofers 5ms "ahead" of the satellites). However, if the delay was added to the woofers, an ad- ditional 45ms of delay (making a total of 55ms) was re- quired for audibility. With a 200Hz crossover, the allow- able additional delays became 9 and 22ms respectively. So, said Poh Ser, by choosing a lower crossover fre- quency, the designer can reduce the audibility of any given amount of delay. Turning to his current subwoofer design, Poh Ser ex- plained that his goal was to reduce both distortion and enclosure size while retaining high acoustic output at low frequencies. The new subwoofer has half the internal volume of his previous design (4 cu. ft. instead of 8), yet has 5dB more output at 3% distortion. Poh Ser's approach was to mount two 12" drivers fac- ing each other at the same end of a cylindrical enclosure. This presents each driver with a "virtual enclosure" which is twice the apparent volume of the real one (each driver "sees" only half the pressure in the enclosure, hence each driver thinks that it is operating in an enclo- sure which is twice as big). Also, by mounting them facing each other, even-order distortion products are mostly eliminated. By choosing drivers with very little odd order-harmonics in the operating range, very low distortion is achieved. On the flip side of the coin, effi- ciency is 3dB lower. Luckily, power is cheap these days, and the woofers used can handle very high power. Like all Poh Ser's designs, this is a vented system (for efficiency and more important, higher maximum output at the lowest frequencies). In a direct comparison, it put out more 16-20Hz than a Hartley 24" woofer, the latter de- veloping a rubbing voice coil during the test). The port is a large diameter (6 inches) tube, with the opening at the bottom. (The drivers are mounted at the top.) Poh Ser commented that most commercial systems use too small a port, resulting in compression and "chuffing" noises due to the extreme velocity of the air in the port. 18-inch woofers are frequently coupled with a 4-inch port (in an $80,000 system, for example). Poh Ser told us that his system is usable from any normal subwoofer crossover frequency—"200Hz, 100Hz, 50Hz, 30Hz"—down to be- low 19Hz, where it is "1 or 2dB down." It still has sub- stantial output at 16Hz. There is less bass boost applied to equalize this sys- tem than was used in his first design. The peak boost oc- curs around 20Hz, and is only 3dB relative to the "high" bass (above 30Hz or so), compared with 6dB boost of the first design. There is also a 12dB/octave infrasonic filter, so the net rolloff rate below cutoff is 36dB/octave. The maximum output he has measured from this particular pair of subwoofers is 116dB at 16Hz with under 3% dis- tortion. This was in a 33'x14'x7.5' basement, the sub- woofers against the middle of the front wall (i.e. about 5 feet from the corners), measured about 10 feet from the speakers, at normal ear level. It was measured by Ira using his Ivie and the CBS test disc. He did not measure the corresponding input power, so we did not have an efficiency figure. Poh Ser did say that the theoretical effi- ciency of the system is about 87dB/watt at 1 meter (87db/watt at 1 meter in half space corresponds to a lit- tle under 2mW acoustic output, i.e., slightly under 0.2 percent efficiency). The woofers have a voice-coil resistance of 5 ohms (at dc). Since they are paralleled, the minimum impedance presented to an amplifier is under 3 ohms (at the port resonance, around 18Hz)—quite low. This results in a fairly high sensitivity. However, if you are worried about the low impedance combined with a possible large phase angle at low frequencies, Poh Ser suggested that the drivers could be wired in series instead of par- allel, raising the minimum impedance to over 10 ohms. "The sensitivity would of course be lower by 6dB." Effi- ciency is clearly unaffected. Poh Ser uses Carver amplifiers at home, by the way (several M1.0ts and an M500t). He used his Carver M500t for driving the subwoofers at this meeting. (It has excellent meters, which allowed us to estimate the power fed to the woofers during the demonstration. The meters were calibrated before being brought to the meeting.) It appeared to drive the under-3-ohm load of the subwoofers without any audible distress, and re- mained quite cool. Each channel—Poh Ser recommends dual sub- woofers—can handle 300 watts long-term average power. He has not yet melted a driver. Demonstrations followed, beginning with a Hin- demith organ selection (on Argo) in which Poh Ser said that the bass was louder (by about 6dB) than the rest of the spectrum (measured using a spectrum analyzer in peak-hold mode and playing through the entire disc). With approximately 150 watts in, Ira Leonard reported that we were getting about 110dB out. "It's almost all around 19Hz, too," Poh Ser told us. When the selection was first played, Al Foster was heard to exclaim, "Wow!" Without the subwoofers and crossover, the dbx 2500s distorted badly on this track, because they were being driven well below their port resonance, which is 1-1/2 octaves higher. Admittedly an unusual case, this is nev- ertheless a good example of what happens to an unpro- tected vented system below resonance, as it becomes un- controlled (Note that in the case of Poh Ser's sub- woofers, the built-in high-pass filter prevents such be- havior by filtering out below-resonance signals). [Also note that single 6" sealed 50Hz-resonance woofers would "distort badly" when overdriven at 19Hz.—DRM] Of course we heard Poh Ser's favorite demo CD, the Philips recording of Saint-Saens' Organ Symphony (Edo